

WAINGAPU MAU HAU, Indonesia

WMO: 973400

Lat: 9.6697S

Lon: 120.2997E

Elev: 12

StdP: 101.18

Time Zone: 8.00 (E08)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	20.2	14.9	10.6	25.4	16.1	11.5	24.4	10.5	29.6	9.1	30.2	2.6	220	0.346

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	7.7	33.5	26.1	33.1	26.3	32.7	26.2	28.0	32.0	27.6	31.7	27.2	31.4	5.6	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.9	22.6	31.1	26.4	21.9	30.7	26.0	21.4	30.4	89.9	32.0	87.8	31.9	86.3	31.6	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.4	7.7	6.8	18.0	35.2	1.1	0.8	17.2	35.8	16.5	36.3	15.9	36.7	15.0	37.3
			15.6	29.2	0.8	1.2	15.0	30.0	14.5	30.7	14.0	31.4	13.4	32.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.4	28.1	27.8	28.0	27.8	27.3	26.4	25.8	25.8	26.7	28.0	28.7	28.5
	DBStd	1.36	0.94	0.85	0.80	0.78	0.98	1.25	1.07	1.07	1.08	0.90	0.72	0.91
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6353	559	497	559	535	537	492	491	490	500	558	561	574
	CDD18.3	3311	301	264	301	285	278	242	233	232	250	299	311	316
	CDH23.3	33998	3135	2603	3090	2995	2768	2162	1976	2057	2518	3469	3710	3514
	CDH26.7	12792	1077	855	1097	1150	1029	729	674	760	997	1502	1579	1341
(14) Wind	WSAvg	2.6	1.9	2.0	2.3	2.4	2.8	2.9	3.1	3.2	3.1	3.0	2.7	2.2
(15) Precipitation	PrecAvg	889	178	180	167	72	43	18	10	4	1	21	45	145
	PrecMax	1438	337	285	336	233	185	219	71	32	26	159	143	276
	PrecMin	481	46	68	25	4	0	0	0	0	0	0	0	34
	PrecStd	246	71	63	72	62	50	42	19	9	5	36	45	64
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.7	33.0	33.2	33.2	32.7	32.1	31.8	32.2	33.0	34.1	34.1	33.8
		MCWB	25.8	26.7	26.9	26.2	25.5	24.8	23.7	23.5	24.5	24.7	26.1	26.9
	2%	DB	32.7	32.3	32.6	32.6	32.1	31.4	31.1	31.5	32.2	33.2	33.4	33.2
		MCWB	26.3	26.5	26.9	26.2	25.3	24.7	23.7	23.4	24.6	25.3	26.4	27.0
	5%	DB	31.9	31.7	32.0	32.0	31.6	30.8	30.5	30.9	31.6	32.6	32.9	32.6
		MCWB	26.4	26.4	26.8	26.1	25.3	24.5	23.5	23.5	24.6	25.5	26.6	26.9
	10%	DB	31.1	30.9	31.3	31.5	30.9	30.2	29.9	30.2	31.0	32.0	32.3	31.8
		MCWB	26.4	26.3	26.5	26.1	25.2	24.3	23.3	23.4	24.6	25.5	26.5	26.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.3	27.8	28.1	27.9	27.5	26.7	26.0	26.1	26.8	27.6	28.4	28.8
		MCDB	31.5	31.4	32.2	31.7	31.4	30.5	29.8	30.5	31.4	32.0	32.5	32.3
	2%	WB	27.5	27.2	27.5	27.3	26.9	25.9	25.2	25.2	26.2	27.0	27.7	28.0
		MCDB	31.2	31.1	31.7	31.3	30.7	30.0	29.5	30.1	30.8	31.8	32.2	32.0
	5%	WB	27.0	26.9	27.1	27.0	26.4	25.4	24.7	24.6	25.7	26.5	27.2	27.4
		MCDB	30.9	30.8	31.2	31.1	30.1	29.5	29.1	29.6	30.3	31.4	31.9	31.5
	10%	WB	26.5	26.4	26.6	26.5	25.9	24.8	24.1	24.1	25.2	26.0	26.8	26.9
		MCDB	30.2	30.2	30.6	30.5	29.5	28.9	28.5	29.2	29.9	31.0	31.4	31.0
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	6.6	7.0	7.5	7.6	8.0	8.8	9.4	9.0	8.5	7.7	6.9
		MCDBR	7.5	7.4	7.6	8.1	8.3	8.2	9.2	9.8	9.5	9.1	8.3	7.7
		MCWBR	3.2	3.4	3.7	3.9	3.6	3.4	4.1	4.5	4.7	4.6	4.1	3.5
	5% WB	MCDBR	6.8	6.9	7.3	7.3	7.2	7.5	8.2	9.1	8.6	8.3	7.8	7.2
		MCWBR	3.5	3.5	3.8	3.9	3.9	3.8	4.3	5.1	4.7	4.5	4.2	3.8
(40) Clear-Sky Solar Irradiance	taub		0.435	0.430	0.417	0.416	0.424	0.417	0.406	0.410	0.435	0.469	0.469	0.451
	taud		2.408	2.423	2.452	2.456	2.424	2.435	2.455	2.422	2.347	2.233	2.270	2.347
	Ebn at Noon		908	909	905	876	838	828	847	869	875	864	871	891
	Edh at Noon		126	124	118	112	110	106	106	114	129	148	143	133
(44) All-Sky Solar Radiation	RadAvg		5.61	5.80	5.82	5.71	5.29	4.91	5.18	5.89	6.53	6.83	6.55	5.84
	RadStd		0.58	0.57	0.49	0.41	0.29	0.25	0.14	0.18	0.29	0.27	0.35	0.40

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page